Programming SQL Server 2019

Working with Tables

Topics

- Table alteration
- Existing column alteration
  - → Changing data type
  - → Changing nullability
- Adding column to existing table
- Dropping a column from existing table
- Computed columns
  - → What is it?
  - → Creating computed columns

Homework

A. Create a table '**Contacts**' with columns as below

Column Name	Data Type	Length	Allow Nulls
ContactID	INT		No
ContactName	VARCHAR	30	No

B. Change the length of ContactName column to 50

C. Add the following columns to the table

Column Name	Data Type	Length	Allow Nulls
ContactTel	VARCHAR	16	Yes
ContactCell	VARCHAR	16	Yes

D. Insert following data into the table

ContactID	ContactName	ContactTel	ContactCell
1	Quazi Ashique		0172889933
2	Hasinur Shahid	8050980	0152980890
3	Mir Mosharaf		0181076543
4	Shaheen Akter		

E. Add the following column to the table

Column Name	Data Type	Length	Allow Nulls	Default value
Address	VARCHAR	100	No	Not available

F. Show me all the rows in the table.

Important Points

-
-
-

Summary

-
-
-

Altering Tables

- Altering a table means modifying a table definition by altering, adding, or dropping columns and constraints, reassigning and rebuilding partitions, or disabling or enabling constraints and triggers.
- For that you have to use, ALTER TABLE statement

Altering an existing column of a table

- Changing an existing column structure has some restrictions
- The following are the restrictions:

→ You cannot change column of TEXT or IMAGE data type
→ You cannot change a computed column
→ You cannot decrease size if value truncates
→ You cannot change a primary key or foreign key

Changing data type or size

→ You can change one data type to another compatible type such as INT to BIGINT; Not CHAR to INT
→ You can lower size if no value loss occurs

- Let's create a table first

```
CREATE TABLE books
(
	id INT not null,
	name VARCHAR (30) NOT NULL,
	author VARCHAR (40) NULL,
	price MONEY NULL
)
GO
INSERT INTO books
VALUES (1, 'SQL', 'Denver', 500)
INSERT INTO books (id, name, price)
VALUES (2, 'C#', 700)
GO
SELECT * FROM books
GO
```

- The output is

id	name	author	price
1	SQL	Denver	500
2	C#	NULL	700

- To alter column the syntax is

```
ALTER TABLE table-name
ALTER COLUMN coumn-name denitions..
```

- Let's alter columns

Try this

```
ALTER TABLE books
ALTER COLUMN name VARCHAR(50) NOT NULL
GO
ALTER TABLE books
ALTER COLUMN id BIGINT NOT NULL
GO
SP_HELP 'books'
GO
```

- You will see the changed column information

→ *Size increase is always possible*
→ *But be careful, decrease in size may lead value loss*

Changing nullability

- NOT NULL to NULL always possible
- NULL to NOT NULL may not be possible, if any row contains NULL in column
- So, price column does not have null value in any row so it is possible to change NULL to NOT NULL for price column
- But author column has null value in second row, so cannot change NULL to NOT NULL for author column

Try this

```
ALTER TABLE books ALTER price money NOT NULL
--OK
GO
ALTER TABLE books ALTER author VARCHAR NOT NULL
-- Error
GO
```

- You are changing nullability but you have to mention data type also

Adding column

- Adding NULL column is always possible
- Adding NOT NULL column you have to provide DEFAULT value
- Syntax is

```
ALTER TABLE table-name
ADD column-name column-definition
```

Try this

```
ALTER TABLE
ADD publishyear INT NULL
ALTER TABLE
ADD edition VARCHAR (30) NOT NULL DEFAULT 'NA'
GO
SELECT * FROM books
GO
```

- The result is

id	name	author	price	publish	edition
1	SQL	Denver	500.00	NULL	NA
2	C#	NULL	700.00	NULL	NA

Dropping a column from a table

- Dropping column has some restrictions

→ You cannot delete a column that has a CHECK constraint. You must first delete the constraint.
→ You cannot delete a column that has PRIMARY KEY or FOREIGN KEY constraints or other dependencies

- Syntax is

```
ALTER TABLE table-name
DROP COLUMN column-name
```

- Let's drop a column

```
ALTER TABLE books DROP COLUMN edition
GO
EXEC SP_HELP 'books'
GO
```

- You will see edition is not in definition

Computed columns

- A computed column is one that is based on values of one or more columns of same table

```
CREATE TABLE products
(
	name VARCHAR (30) NOT NULL,
	regularprice MONEY NOT NULL,
	discount DECIMAL(4, 2) NOT NULL,
	currentprice AS regularprice*(1-discount)
)
GO
```

- You cannot insert values for computed column, they automatically calculated

Try this

```
INSERT INTO products (name, regularprice, discount)
VALUES ('P1', 500.00, .10)
INSERT INTO products (name, regularprice, discount)
VALUES ('P2', 900.00, .15)
GO
SELECT * FROM products
GO
```

- The output is

name	regularprice	discount	currentprice
P1	500.00	0.10	450.00
P2	900.00	0.15	765.00